SYNOPSIS (ES 451)

ON

ROUTE SHARE : AN INTELLIGENT CARPOOL SYSTEM

Submitted for partial fulfilment of award of the degree of

Bachelor of Technology

In

Computer Science & Engineering

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Introduction

The proposed project RouteShare is an intelligent, AI-powered carpooling platform designed to optimise urban transportation by connecting nearby riders with drivers going along similar routes. The system leverages machine learning, real-time geolocation tracking, route optimisation algorithms, and secure digital payments to deliver a safe, efficient, and cost-effective ride-sharing experience.

Problem Statement

Transportation Inefficiency:
Urban areas face increasing traffic congestion, fuel consumption, and travel delays due to the rise in private vehicles.

Lack of Intelligent Ride Matching:
Traditional ride-sharing systems often fail to match riders and drivers based on the optimal route, distance, and time.

Safety & Trust Concerns:
Passengers face uncertainty regarding driver identity, route transparency, and secure payment handling.

Manual Booking Limitations:
Non-automated systems require manual coordination, creating delays, miscommunication, and inefficiencies.

Ineffective Cost Management:
Passengers and drivers lack a transparent structure for fare calculation and payment settlement.

Objectives

Primary Objectives

To develop an **AI-powered carpool system** capable of smart ride matching based on proximity, route similarity, and travel timing.

To implement **real-time tracking** using maps, geolocation APIs, and WebSocket-based updates.

To ensure **safe and secure authentication** using JWT and encrypted user data.

To enable **cashless transactions** through Stripe payment integration.

To design an intuitive, responsive **React-based user interface** for drivers and passengers.

To build an efficient backend using **Node.js, Express, and MongoDB Atlas.**

Secondary Objectives

To optimise routes using **OpenRouteService** for reduced travel distance and time.

To provide AI-based explanations of route optimisation using **HuggingFace Mistral** models.

To introduce **rating, feedback, and verification** features that promote user trust.

To enable efficient ride management with notifications, **seat tracking, and booking history.**

To ensure scalability for future enhancements such as **corporate pooling and dynamic surge pricing.**

Feasibility Study

Technical Feasibility

Real-time Tracking: Achieved through Socket.io and geolocation services

AI Route Optimisation: OpenRouteService + ML models for intelligent matching

Secure Payments: Stripe API for safe transactions

Scalable Architecture: MERN stack ensures modular expansi

Need and Significance

Environmental Impact

Reduced carbon emissions by promoting shared mobility.

Cost Efficiency

Lower travel cost for passengers and increased utilisation for drivers.

Urban Mobility Enhancement

Helps reduce traffic congestion, fuel consumption, and road load.

Improved Safety and Transparency

Secure authentication, verified profiles, and digital payments ensure trustworthiness.

Productivity Enhancement

Automated ride matching and real-time tracking save time and improve user convenience.

Proposed Methodology in Brief

User Authentication Module:
Login/Signup using JWT, encrypted passwords, and verification emails.

Ride Creation Module:
Drivers enter route, timing, available seats, and price.

AI-Powered Ride Matching:
Uses OpenRouteService + custom ML logic to match riders efficiently.

Real-Time Tracking:
Allows users to track ride progress via Leaflet maps and Socket.io.

Payment Gateway:
Integrates Stripe for secure online payments.

AI Explanations:

Generates human-like route explanations using HuggingFace Mistral 8×7B.

Database Management:
MongoDB Atlas stores ride data, profiles, payment logs, and notifications.

Implementation

Phase 1: Foundation

- Setup of MERN stack, backend APIs, and database structure
- User authentication with JWT
- Basic ride creation and profile modules

Phase 2: Core Development

- Implement AI-based ride-matching
- Configure OpenRouteService for route optimisation
- Integrate Stripe payments
- Develop passenger ride booking and driver ride-managing modules

Phase 3: Integration & Testing

- Add real-time location tracking
- End-to-end testing with actual routes
- UI refinement and responsive design

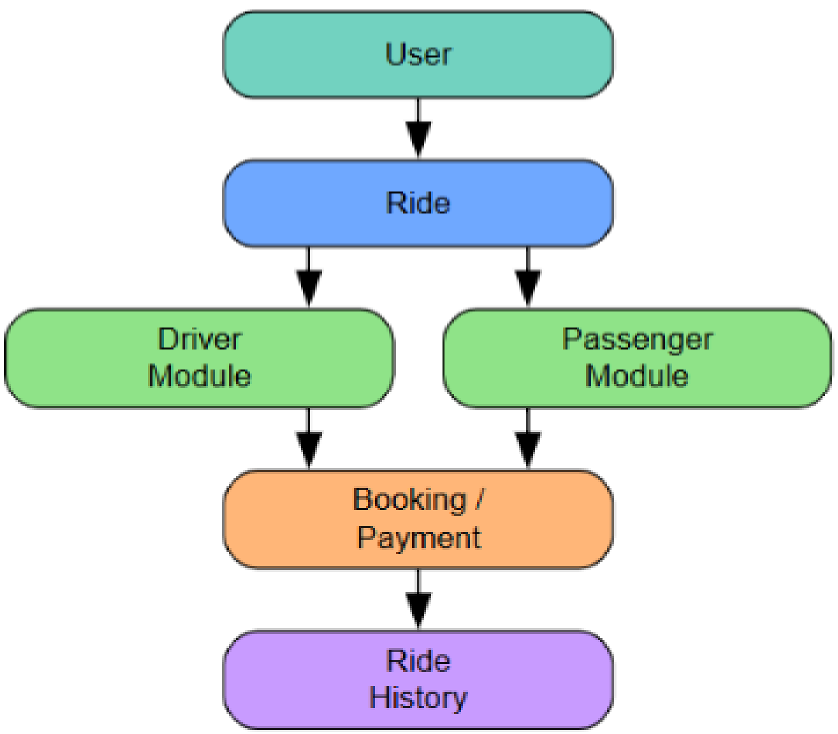
Phase 4: Deployment & Refinement

- Hosting backend, frontend, and database
- Usage monitoring, bug fixes, and performance improvements
- Addition of advanced features such as analytics and surge pricing

Expected Outcomes and Benefits

- **Faster Travel:** Optimised routes reduce commute time.
- **Reduced Cost:** Shared rides lower per-person travel expenses.
- **Eco-Friendly:** Fewer vehicles on road → lower pollution.
- **Complete Transparency:** Real-time maps & digital payments.
- **Improved User Trust:** Verified profiles and streamlined process.
- **Scalable Solution:** Supports large user base with cloud integration.

Framework Architecture Overview



Hardware Requirements

Development Environment

Minimum Requirements

- **CPU:** Intel i5-8400 or AMD Ryzen 5 2600 (6 cores)
- **RAM:** 16 GB DDR4
- **Storage:** 500 GB SSD
- **GPU:** Intel® Iris®

Recommended Requirements

- **CPU:** Intel i7-10700K or AMD Ryzen 7 3700X (8+ cores)
- **RAM:** 32 GB DDR4
- **Storage:** 1 TB NVMe SSD
- **GPU:** Intel® Iris®

Software Requirements

Development Stack

Backend Framework

- **Node.js 18+** with **Express.js** for backend API development
- **MongoDB Atlas** for cloud-based data persistence
- **Mongoose** for schema modelling
- **JWT (JSON Web Tokens)** for secure authentication
- **Socket.io** for real-time ride tracking and updates
- **Stripe API** for secure online payments
- **OpenRouteService API** for route optimisation and distance calculation

AI/ML & Utility Libraries

- **Hugging Face Mistral 8×7B API** for AI-based explanations

- **Bcrypt.js** for password hashing
- **OpenRouteService JavaScript SDK** for geolocation and routing

Frontend Technologies

- **React 18+** (Vite-based setup) for dynamic and responsive UI
- **Tailwind CSS** for modern, utility-first styling
- **Leaflet.js** for interactive maps and real-time location tracking
- **React Router** for frontend routing
- **Stripe.js** and **React-Stripe-Elements** for payment handling
- **React Icons** & **ShadCN Components** for UI elements

Development Tools (Version Control & CI/CD)

- **Git & GitHub** for source control
- **VS Code** for development
- **Postman** for API testing

Scope

The system aims to modernise urban mobility by providing a reliable AI-driven carpooling platform. It serves **daily commuters, students, corporate employees, and city travellers**, offering a smart, affordable, and eco-friendly transportation alternative. Future enhancements may include advanced predictive models, dynamic fare computation, corporate ride pooling, and integration with public transportation systems.

References

[1] OpenRouteService Documentation – Route Optimisation and Geolocation APIs

[2] Stripe Developer Docs – Payment Integration and Security Guidelines

[3] MongoDB Atlas Documentation – Cloud Database Management

[4] Hugging Face – Mistral AI Model Documentation

[5] IEEE & ACM Digital Library – Research on AI-Driven Transportation Systems